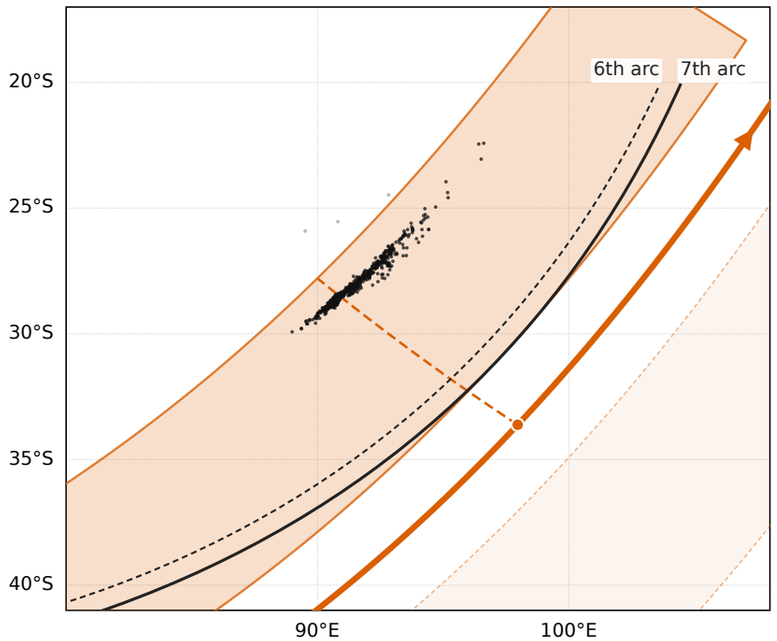
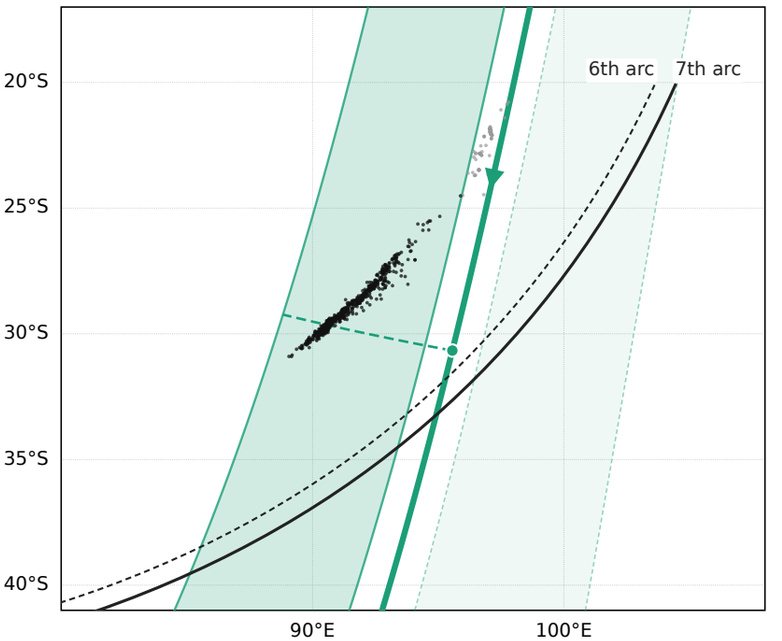
Where four radar satellites could have imaged MH370 in flight, 7–8 March 2014

One panel per satellite, in time order. Each radar images a strip beside its path, not below it. It can point to either side, but to one side at a time.

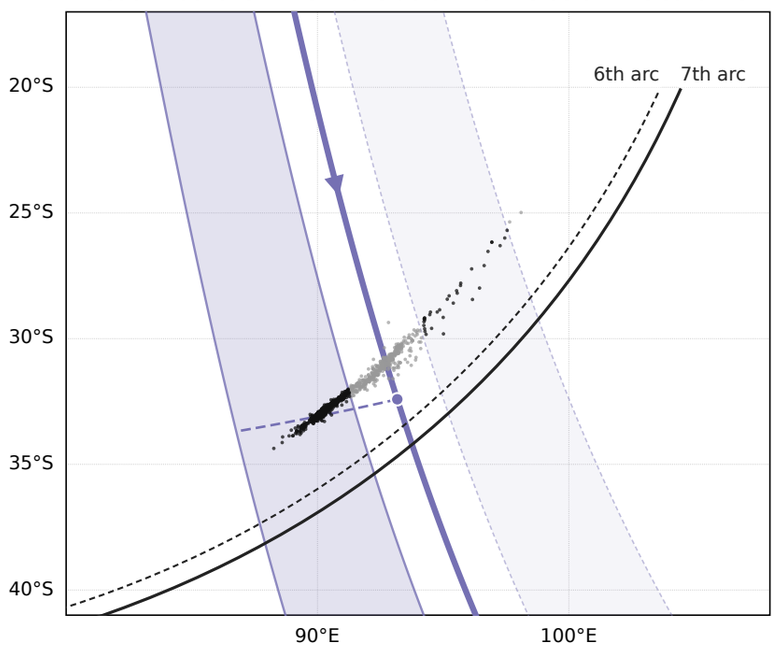
a Lacrosse 5, 23:12 UTC: aircraft on its left, P inside reach 1.00
reach 236–999 km either side of the line below the satellite



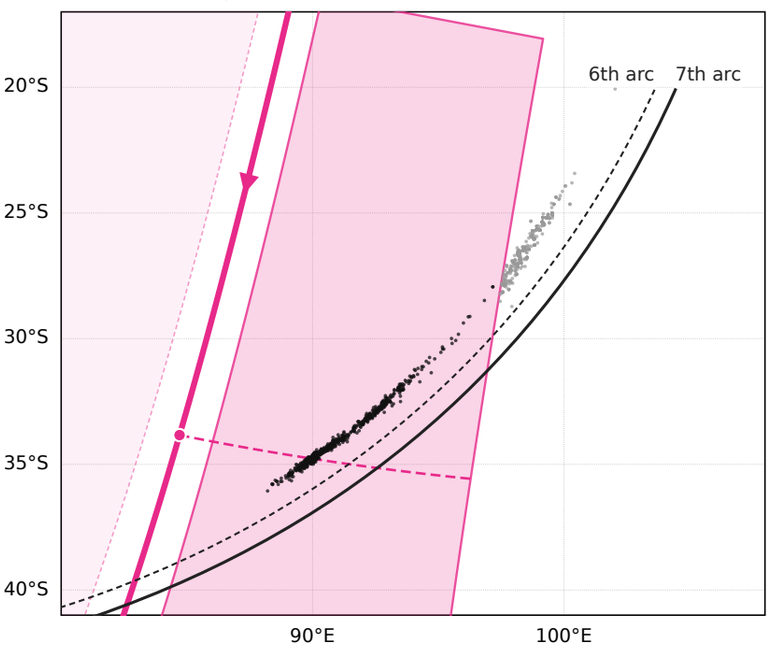
b RISAT-1, 23:20 UTC: aircraft on its right, P inside reach 1.00
reach 107–670 km either side of the line below the satellite



c Kondor (Kosmos 2487), 23:45 UTC: aircraft on its right, P inside reach 0.81
reach 166–615 km either side of the line below the satellite



d RADARSAT-2, 23:59 UTC: aircraft on its left, P inside reach 0.99
reach 125–1,078 km either side of the line below the satellite



Path of the point below the satellite, ±5 min (arrow: direction of travel)
Radar reach on the side where the aircraft was
Radar reach on the other side (one side per pass)
Line across the track at the moment of the sweep
Posterior aircraft positions inside the reach
Posterior aircraft positions outside the reach
7th arc (00:19)
6th arc (00:11)

Plain-language note: in each panel the black dots show where the aircraft probably was when that satellite passed. Where they lie in a shaded strip, the satellite could have imaged the aircraft if it had been ordered to point to that side. Nothing here shows that an image was taken.

Technical: core run J routes.npy, waypoints family (48,000 routes, P(family) 0.9982), linear interpolation along 10-min chords (about 25 NM tolerance at turns); up to 1,200 in-reach samples drawn per panel. SGP4 on March 2014 element sets, 5 s steps; zero-Doppler crossing per route; reach = ground offsets at the minimum and maximum incidence, either side (RADARSAT-2 10–60°, RISAT-1 12–55°, grade B; Lacrosse 5 20–60°, Kosmos 2487 20–55°, grade C; Lacrosse look side not published, both assumed). P inside reach is the waypoints-family share; four-family mixture: 1.00, 1.00, 0.80, 0.99. Kosmos 2487: 3 % of in-reach routes lie on its left. Cross-track line at the median sweep time; times rounded to the minute. FL350 assumed; the radar would place the aircraft 7–18 km toward itself and 2–18 km along its track.